

Research Assessment

Wind Speed Time-Series Forecasting

Dataset	windspeed.csv
Task	Benchmark forecasting models and test robustness against Gaussian noise.
Deadline	22/6 at 11:59 PM
Evaluation metrics	MAE and RMSE

Assessment Objective

You are required to build and evaluate forecasting models using the provided wind speed time-series dataset. The goal is to compare classical machine learning and deep learning models, then analyze the robustness of the best model under noisy conditions.

Required Work

- Use the provided **windspeed.csv** dataset.
- Perform necessary preprocessing, including handling missing values if present, scaling or normalization if needed, and a proper train/test split.
- Benchmark **2 machine learning models** for forecasting.
- Benchmark **2 deep learning models** for forecasting.
- Evaluate all four models using **MAE** and **RMSE**.
- Select the best-performing model based on the evaluation results and clearly justify the selection.

Noise Robustness Test

After selecting the best model, test its robustness by injecting **Gaussian noise** into the test data and evaluating performance again.

Noise Type	Noise Levels	Metrics
Gaussian noise	5%, 10%, and 20%	MAE and RMSE

Expected Deliverables

- A clean, runnable notebook or Python script.
- A short report containing: preprocessing steps, models used, results tables, best-model justification, robustness analysis, limitations, and possible improvements.
- Tables reporting MAE and RMSE for all benchmarked models and for each Gaussian-noise level.
- Plots comparing actual vs predicted wind speed values are encouraged.

Submission Notes

- The code should be reproducible and clearly organized.
- Candidates should avoid data leakage and keep the test set unseen during training and model selection.
- Any assumptions made during preprocessing, model selection, or noise injection should be briefly explained.